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Assignment No. 1: Requirement Analysis and User Story Writing

PROJECT OVERVIEW:

The client wants an online bookstore where users can browse and search for books, buy books online, manage their account and have an overall safe and easy shopping experience.

REQUIREMENT ANALYSIS:

User Preferences:

User Need	Reason
They want to be able to find books by typing their titles, author name or category.	It saves time.
They want to be able to see all details for a book including description, price, author name, cover image, rating, and availability.	It helps them make a better decision of whether they want the book or not.
They want to be able to filter books based on price range, language, publication year, and rating.	It helps them narrow down their choices.
They want to be able to sort books in ascending and descending order based on prices.	It allows them to find books that fit their budget.
They want to be able to add books to their “wishlist”.	It helps them find books that they like later.
They want to be able to read the reviews from other buyers.	It allows them to trust their purchase more.
They want to be able to pay safely on the site using their credit, debit, and visa card.	It makes them feel secure.
They want to be able to add items to their cart and see their grand total.	It helps them get an overview of all products along with the prices before the final purchase.
They want to be able to track their order after purchase.	It helps them know when

	their order will arrive.
They want the app to save their address and payment information.	This makes it easy for them to place orders in the future.
They want the app to save their order history.	This helps them in case they want to purchase the same books again.
They want to get an email confirmation after their order.	This provides them with a proof of order.
They want to be able to use the website on mobiles, tablets, and laptops.	This allows them to shop from anywhere.

Functional Requirements:

1. **FR-1:** The system enables users to perform book searches through three search criteria which include title, author name and category.
2. **FR-2:** The system provides a complete book display which contains description, price, author name, cover image, rating and availability information.
3. **FR-3:** The system provides users with multiple filters which enable them to search by price range, language, publication year and rating.
4. **FR-4:** The system enables users to arrange books according to their price which includes both ascending price order and descending price order.
5. **FR-5:** The system enables users to create their wishlist by saving books which they want to examine at a future time.
6. **FR-6:** The system presents user-generated reviews together with ratings for every book displayed on its respective page.
7. **FR-7:** The system needs to implement a protected payment system which accepts credit cards, debit cards and visa cards for transactions.
8. **FR-8:** The system enables users to create their shopping cart by adding products to the cart and will show all items that are present in the cart along with the total amount.
9. **FR-9:** The system enables users to track their orders through the system which displays their order status including processing, shipping and delivered.
10. **FR-10:** The system enables users to enter their shipping address and payment details which get saved in their account profile.

11. **FR-11:** The system retains every order record which belongs to each user account.
12. **FR-12:** The system sends an email to users which confirms their order after they complete their successful purchase.
13. **FR-13:** The system adapts its display to function properly on all display dimensions.

Non-Functional Requirements:

1. **NFR-1:** The system requires that all passwords should be stored through encrypted hashing methods.
2. **NFR-2:** The system needs to protect payment details through a payment gateway which meets PCI DSS security standards.
3. **NFR-3:** The user session must end after 1 hour of no activity.
4. **NFR-4:** All search results must complete their loading process within 3 seconds.
5. **NFR-5:** The book detail page must load within 3 seconds.
6. **NFR-6:** The checkout process must finish its operations within 10 seconds.
7. **NFR-7:** The website must maintain 99.9% availability throughout the entire operating period.
8. **NFR-8:** The website must maintain complete responsiveness across mobile devices, tablet devices, and laptop devices.
9. **NFR-9:** The website font size should remain legible across all types of devices.
10. **NFR-10:** All website buttons need to work as tappable elements across all types of devices.
11. **NFR-11:** Users should receive their order confirmation email within 5 minutes after completing their purchase.
12. **NFR-12:** The system must show real-time updates about the current status of the order.
13. **NFR-13:** The website must support simultaneous access by 1000 or more users during times of maximum user demand.

USER STORIES:

Account Management:

1. **US-1:** As a new user, **I want** to be able to create an account and login successfully into the site using the account, **so that** I can manage all my data related to the site.

2. **US-2: As a user with an existing account, I want to be able to log into the site using my existing credentials, so that I can access information stored in my profile.**

Search & Discovery:

1. **US-3: As a busy shopper, I want to search books by title, author name, or category, so that I can find the book that I need without browsing through hundreds of books.**
2. **US-4: As a budget-conscious buyer, I want to filter books by price range, language, publication year, and rating, so that I can find books that match my preferences.**
3. **US-5: As a smart-buyer, I want to sort books in ascending and descending order, so that I can find books that fit by budget.**

Book Information:

1. **US-6: As a curious buyer, I want a detailed book page with information about the book including description, price, author name, cover image, rating, and availability, so that I can decide whether I want to buy the book or not.**
2. **US-7: As a cautious buyer, I want to see the reviews and ratings of other users, so that I can trust my purchase before spending money.**

Wishlist:

1. **US-8: As a shopper who likes to compare, I want to be able to save items in a wishlist, so that I can find them easily when I am ready to buy.**

Shopping Cart:

1. **US-9: As a shopper buying multiple items, I want to add items to my cart, so that I can purchase several books in one order.**
2. **US-10: As a careful shopper, I want to see the grand total of items in my cart, so that I know the amount I am paying before I complete the purchase.**

Checkout & Payment:

1. **US-11: As a security conscious customer, I want to pay for my order using my credit, debit or visa card using a secure payment method, so that my card information is not misused.**
2. **US-12: As a frequent buyer, I want to store my address and payment information on the site, so that I don't have to enter it again and again.**

3. **US-13: As a buyer, I want** to receive an order confirmation email after placing the order, **so that** I have proof of my order.

Order Management:

1. **US-14: As a buyer, I want** to track my order status (processing, shipped, delivered), so that I know when my order will arrive.
2. **US-15: As a frequent buyer, I want** to see my order history when I return to the website, **so that** I can place an order for the same items again if I want.

Device Support:

1. **US-16: As a user that uses multiple devices to shop, I want** to be able to use the website on mobile, tablet and laptop, **so that** I can shop from anywhere, anytime.

Security:

1. **US-17: As a user who is conscious about my privacy, I want** my password to be stored in encrypted format, **so that** no one steals my account information.
2. **US-18: As a user, who sometimes steps away from my computer, I want** my session to timeout after 1 hour of inactivity, **so that** no one else can use it while I am away.

Performance:

1. **US-19: As a shopper in a hurry, I want** the website search results and book detail page to load in 3 seconds, **so that** my time is not wasted.
2. **US-20: As a customer visiting the site during peak hours, I want** the website to handle a large number of users, **so that** I can purchase books during sales.

ACCEPTANCE CRITERIA:

Feature: User Account Management

Background: User is on the bookstore website

Scenario: User creates a new account

Given user is on the sign-up page

When the user enter valid email address

And the user enters valid password

And the user clicks the “create account” button

Then the account gets created successfully

And user is automatically logged-in

And the user is redirected to the dashboard page.

Scenario: User logs in using an existing account

Given user is on the login page

When the user enters an email address of an existing account

And user enters the correct password of an existing account

And the email and password match

And the user clicks the “login” button

Then the user is successfully logged-in

And the user is redirected to the dashboard page.

Scenario: Session timeout after inactivity

Given user is logged in

When user does no activity for 1 hour

Then they get logged out

And they see the message that your session has expired, please log in again

Feature: Search, filter and sort

Background:

Given user is on the website home page

And the site has some books in the catalog

Scenario: Search books by title and author

When user types a book title like “Harry Potter” in the search bar

And user presses search button

Then the search results are visible within 3 seconds

And all the results show title, author, price and cover image

Scenario: Filter books based on price and rating

Given the user has performed some search

When user selects the price range Rs. 1000 - Rs.2000

And they select the rating of “4 stars and above”

Then only the books that match the criteria are visible.

Scenario: Sort books based on price

Given user has performed some search

When user selects sorting from low to high

Then the cheapest book is shown first

And the most expensive book is shown at the end.

Feature: Book Information & Reviews

Background: User is on the book store website

Scenario: User view details of a book

When user clicks on any book card

Then the book detail page loads within 3 seconds

And it shows the book description, price, author name, cover image, rating, and availability

Scenario: User views reviews of a book

Given user is on the book detail page

When user scrolls down to reviews section

Then they can see all reviews of the book along with their text and rating.

Feature: Wishlist

Background: Given user is logged in the website

Scenario: User adds items to the wishlist

Given the user on book detail page

When they click the “Add to wishlist” button

Then the item gets saved to their wishlist

And the button text changes to “Remove from wishlist”

Scenario: User views items of the wishlist

Given user has some items in their wishlist

When user clicks on “My wishlist” button

Then they are redirected to the Wishlist page

And they are shown all the books that are present in their wishlist

And when they click on any wishlist item they are redirected to the book detail page of that book.

Feature: Shopping Cart

Background: User is logged in our website

Scenario: Users adds items to the cart

Given user is on the book detail page

When user clicks the “Add to cart” button

Then the item is added to cart

And the item count on cart icon is updated

Scenario: User views items in the cart

Given user has some items in the cart

When user click on the cart icon

Then they are redirected to the cart page

And they can see all the items in the cart along with the grand total

Scenario: User removes an item from cart

Given user has some items in the cart

And user is on the cart page

When the user clicks on the “Remove” button that is present with the item

Then the item gets removed from the cart

And the grand total gets updated

Feature: Checkout & Payment

Background:

Given the user is logged in

And they have some items in the cart

Scenario: User places order successfully

Given the user is on the cart page

When the user clicks the “Proceed to checkout” button

And they enter or select saved address

And they enter or select saved payment method and information

And they click “Place order” button

Then the order is successfully placed

And the overall checkout process is completed within 10 seconds

Scenario: Payment is processed securely

Given user is on the payment page

When user enters their credit/debit/visa card information

Then the payment is processed through a secure payment gateway that is PCI DSS compliant

And user payment information is not stored in plain text

Scenario: User receives an order confirmation email

Given user has placed an order successfully

Then they receive an order confirmation email within 5 minutes

And the email will contain order number, items, and total amount

Feature: Order Management

Scenario: User can view the current status of their order

Given user has placed an order successfully

When the go on “My Orders” page

Then they can see all their order along with status of their current order

And the order status updates in real-time

Scenario: User can view they order history

Given user has some past orders

When they go on “My Orders” page

They can see all their current and past orders along with their dates, items and total amount.

Feature: Device Support

Background: Given user wants to access our site

Scenario: User tries to access our site on all devices

When user tries to open our site on mobile phone

And they try to open same site on tablet

And they try to open same on laptop

Then the website in responsive on all screen sizes

And buttons are tappable on all devices

And font size is readable on all devices

SPRINT PLANNING:

Each Sprint Length: 1 week

Sprint 1:

Goal: Users will be able to create accounts through login and signup processes

Selected User Stories: US-1, US-2, US-18

Tasks: Create design for login and signup interfaces, develop system for user registration, conduct email address and password verification, execute user login access control, develop system for handling user sessions, establish system to terminate user sessions after one hour of inactivity.

Sprint 2:

Goal: Users will be able to search and apply filters and sorting features

Selected User Stories: US-3, US-4, US-5, US-19

Tasks: Create design for search bar, establish filter and sort options through dropdown menu , develop system to search using title, author and category, develop system to filter results according to price, language, publication year and rating, develop system to sort results according to price starting from lowest to highest and highest to lowest, enhance search system to present results within three seconds.

Sprint 3:

Goal: Users will be able to view book details and read book reviews

Selected User Stories: US-6, US-7

Tasks: Create design for book detail page, display book details which include description, price , author, image, rating and availability, establish reviews section on this page, display all user reviews, make the page load within three seconds.

Sprint 4:

Goal: Users will be able to manage items through their wishlist and cart features

Selected User Stories: US-8, US-9, US-10

Tasks: Develop system that enables users to add and delete items from their wishlist, create the wishlist page, develop system that allows users to add items to their shopping cart, create design for cart page, present shopping cart products and total cost on cart page, develop system that allows users to delete items from their shopping cart.

Sprint 5:

Goal: Users will be able to complete their purchase through payment and checkout process

Selected User Stories: US-11, US-12, US-13

Tasks: Create design for checkout page, develop functionality to store user address details and payment information, connect to a payment gateway that meets PCI DSS standards, create system that enables customers to place their orders, develop system that sends order confirmation emails

to customers within five minutes, enhance the checkout system to finish all tasks within ten seconds.

Sprint 6:

Goal: Implement order management

Selected User Stories: US-14, US-15

Tasks: Create “My orders” page, show user list of all these past and current orders, display current status of orders to user (processing, shipped, delivered), update order status in real-time.

Sprint 7:

Goal: Implement responsiveness and improve performance along with security

Selected User Stories: US-16, US-17, US-20

Tasks: Make website responsive on various screen sizes, make buttons tappable and fonts readable on all screen sizes, add password encryption through hashing, make sure our website can handle 1000+ concurrent users, make sure the website has 99.9% uptime, perform performance testing on site and fix the bugs.

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